

**Exercise 1:** Generate matrices A, with random Gaussian entries, B, a Toeplitz matrix, where  $A \in \mathbb{R}^{n \times m}$  and  $B \in \mathbb{R}^{m \times n}$ , for  $n = 200$ ,  $m = 500$ .

**Exercise 2:** Matrix operations

Calculate  $A + A$ ,  $AA^T$ ,  $A^T A$  and  $AB$ . Write a function that computes  $A(B - \lambda I)$  for any  $\lambda$

**Exercise 3:** Solving a linear system

Generate a vector b with m entries and solve  $Bx = b$ .

**Exercise 4:** Norms

Compute the Frobenius norm of A:  $\|A\|_F$  and the infinity norm of B:  $\|A\|_\infty$

**Exercise 5:** Nearest neighbor

Write a function that takes a value Z and an array A and finds the element in A that is closest to Z. The function should return the closest value, not index.

Hint: Use the built-in functionality of Numpy rather than writing code to find this value manually